Local projections: Simultaneous inference, shrinkage, bias correction, bootstrap

Mikkel Plagborg-Møller

University of Chicago

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Outline

1. Simultaneous inference
2. Shrinkage
3. Bias correction
4. Bootstrap

Simultaneous inference

- We often want to test hypotheses involving impulse responses at multiple horizons. Called simultanous or joint testing.
- We can't do this based on the pointwise confidence band that we have reported so far. The constituent confidence intervals only guarantee 68% coverage (say) at each horizon *separately*.
  - Suppose we rejected the null that *all* impulse responses equal 0 if *any* of the pointwise confidence intervals exclude 0. Then we would have high likelihood of rejecting when the number of horizons is large due to random noise.
- If we want to test a specific hypothesis involving multiple LP coefficients, we need to first "stack" the several LP regressions with Stata's `suest` command. This takes into account the correlation of estimates across horizons. Then we can use the `test` command.

```
postfile f h ir se using lp, replace
forvalues h=0/‘maxhorz’ {
  qui reg F‘h’.‘outcome’ ‘impulse’ ‘controls’, robust
  post f (‘h’) (_b["‘impulse’"]) (_se["‘impulse’"]) // Store
  qui reg F‘h’.‘outcome’ ‘impulse’ ‘controls’
  estimates store lp‘h’ // Store regression results for later joint inference
}
postclose f

suest lp*, vce(robust) // Stack regression results

local teststr ""
forvalues h=0/‘maxhorz’ { // Build string of LP coefficients to be tested
  local teststr "‘teststr’ [lp‘h’_mean]‘impulse’"
}
test ‘teststr’, matvlc(vcov_matrix) // Test null of no effect at any horizon
```

```
. test `teststr', matvlc(vcov_matrix) // Test null of no effect at any horizon

 ( 1)  [lp0_mean]gtmp_bkly_aw_dtfe2s = 0
 ( 2)  [lp1_mean]gtmp_bkly_aw_dtfe2s = 0
 ( 3)  [lp2_mean]gtmp_bkly_aw_dtfe2s = 0
 ( 4)  [lp3_mean]gtmp_bkly_aw_dtfe2s = 0
 ( 5)  [lp4_mean]gtmp_bkly_aw_dtfe2s = 0
 ( 6)  [lp5_mean]gtmp_bkly_aw_dtfe2s = 0
 ( 7)  [lp6_mean]gtmp_bkly_aw_dtfe2s = 0
 ( 8)  [lp7_mean]gtmp_bkly_aw_dtfe2s = 0
 ( 9)  [lp8_mean]gtmp_bkly_aw_dtfe2s = 0
 (10)  [lp9_mean]gtmp_bkly_aw_dtfe2s = 0
 (11)  [lp10_mean]gtmp_bkly_aw_dtfe2s = 0

           chi2( 11) =   30.32
         Prob > chi2 =    0.0014
```

Simultaneous confidence band

- It is also desirable to have a graphical way to visualize the uncertainty about the entire impulse response *function* (IRF), not just the individual horizons.
- A simultaneous confidence band covers the *entire* IRF with probability 68% (say).
- It can be constructed by enlarging the usual pointwise critical value. One could do a Bonferroni adjustment, but it's more efficient to report the sup-t band:

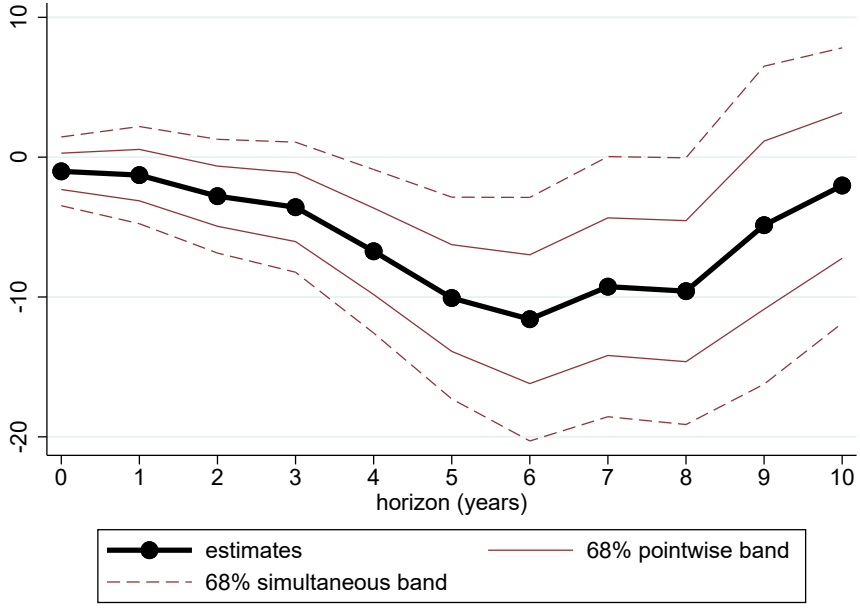
$$\texttt{estimate} \pm \texttt{cv} * \texttt{SE}.$$

The critical value cv is computed based on the correlation matrix of the LP estimates across all horizons.

```
* Critical value for simultaneous confidence band
estimate_supt_critical_value, vcov_matrix(vcov_matrix) conf_level(0.68)
local supt_cv = r(critical_value)

use lp, clear
gen lower_pw = ir-se
gen upper_pw = ir+se
gen lower_si = ir-‘supt_cv’*se
gen upper_si = ir+‘supt_cv’*se
```

Note: `estimate_supt_critical_value` command downloaded from `https://github.com/ryanedmundkessler/simultaneous_confidence_bands`


10
0
-10
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6
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10
horizon (years)
estimates
68% pointwise band
68% simultaneous band

Simultaneous confidence band

- We can use the simultaneous confidence band to visually test a great variety of shape restrictions on the IRF.
- Simply check whether some function with the hypothesized shape is contained in the simultaneous band.
- Examples:
  - Hypothesis: responses at all horizons equal 0 $\Rightarrow$ reject if any part of the band excludes 0.
  - Hypothesis: the IRF is monotonically increasing in h $\Rightarrow$ reject if we can't draw any monotonic function inside the band
  - Hypothesis: the response at $h = 0$ is larger than the response at $h = 4$ $\Rightarrow$ reject if the upper band at $h = 0$ is entirely below the lower band at $h = 4$

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Smoothing out the IRF

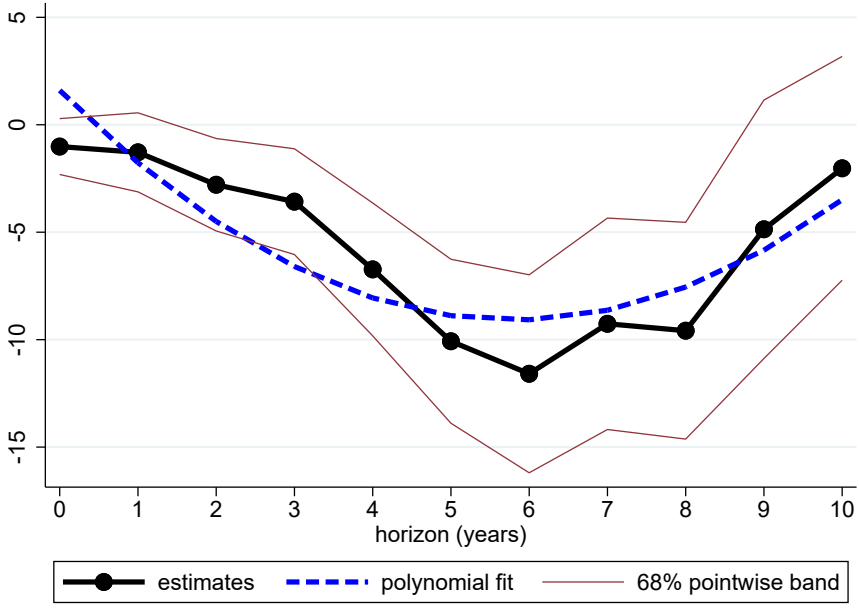
- In some applications the LP-estimated IRF bounces around erratically from horizon to horizon. Unlike VARs with short lag lengths, there is nothing constraining the LP IRF to be smooth.
- In some sense, this is good: we have seen that the LP estimator prioritizes low bias at the expense of higher variance. The jaggedness comes from the latter.
- But since we often believe that the true IRF is fairly smooth, we could consider reporting a smoothed-out estimate in addition to the original one.

Parametric fit

- One simple approach is to fit a parametric functional form $f(h)$ to the LP impulse responses $\hat{\theta}_h$ for $h = 0, 1, \ldots, h_{\max}$.
- For example, a 2nd-order polynomial: $f(h) = a_0 + a_1 \times h + a_2 \times h^2$.
  - If all estimated impulse responses are positive, we could alternatively fit a polynomial to the logarithm of the estimated responses. Jordà & Taylor (2025) recommend this option, if possible. (If all responses are negative, just flip their sign.)
  - We could consider higher-order polynomials to fit more complicated shapes. But the higher the order, the less smoothness.
- We should still report the usual pointwise or simultaneous LP confidence band. The point of the parametric fit is only as a visual aid; we don't believe the true IRF is *exactly* a polynomial, so we won't report a confidence band that assumes this.

```
use lp, clear

gen h2 = h^2
local fit_reg h h2 // Covariates in parametric fit
reg ir `fit_reg'
predict ir_fit // Best-fitting polynomial
local sum_sq_res = e(rss) // Sum of squared residuals, for later
```


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horizon (years)
estimates
polynomial fit
68% pointwise band

Shrinkage

- The polynomial fit imposes a particular functional form regardless of how well it fits the original LP estimates, and regardless of how precise/noisy those estimates are.
- A more sophisticated approach is to shrink the original LP estimates part of the way toward the restricted fit.
- The degree of shrinkage should ideally depend on the data:
  - If the goodness-of-fit of the polynomial specification is poor, shrink less.
  - If the original estimates are noisier (larger SE), shrink more.
- Barnichon & Brownlees (2019) propose a smooth LP estimator that adds a penalty term to the linear regression, such that the estimator favors smoother IRFs. They select the magnitude of the penalty in a data-driven way.

Shrinkage

- Barnichon & Brownlees provide Matlab and R code for their estimator.
- I will instead consider a simpler smooth IRF estimator proposed in my PhD dissertation (Plagborg-Møller, 2016). It has two advantages:
  - Easy to implement in Stata.
  - Can be applied to any LP setting (e.g., IV, panel data), as it just requires the original LP estimates and their var-cov matrix.
- The shrinkage estimator is computed as

$$\hat{\theta}_h^{\text{smooth}} = (1 - c) \times \hat{\theta}_h^{\text{original}} + c \times \hat{\theta}_h^{\text{fit}}, \quad h = 0, 1, \ldots, h_{\max},$$

where the shrinkage factor $c \in [0, 1]$ depends on the ratio of noise to goodness-of-fit.

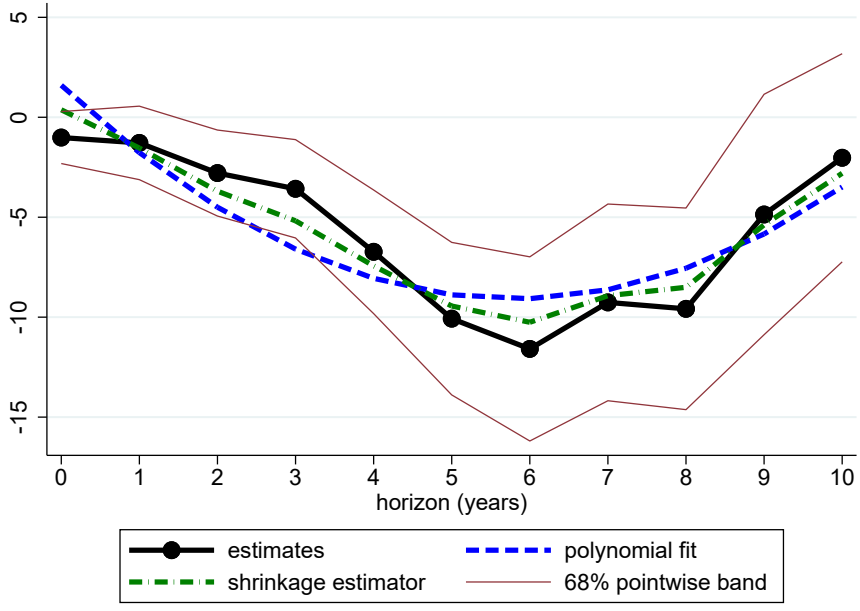
```
* Compute trace of var-cov matrix of residuals
vmat vcov_matrix, names(mat) // Make columns of var-cov matrix into variables
local trace = 0
forvalues h=0/`maxhorz' {
  local i = `h'+1
  qui reg mat`i' `fit_reg'
  predict mat`i'_res, resid
  local trace = `trace' + mat`i'_res[`i']
}

* Shrinkage
local shrink = min(`trace'/`sum_sq_res', 1) // Shrinkage factor
gen ir_shrink = (1-`shrink')*ir + `shrink'*ir_fit // Shrinkage estimator
```

```
. disp as text "Shrinkage factor = " as result `shrink'
Shrinkage factor = .53063164

. list h ir ir_fit ir_shrink
```

	h	ir	ir_fit	ir_shrink
1.	0	-1.01357	1.601302	.373964
2.	1	-1.282557	-1.766313	-1.539253
3.	2	-2.790329	-4.498682	-3.696836
4.	3	-3.577711	-6.595806	-5.179208
5.	4	-6.729412	-8.057682	-7.434234
6.	5	-10.07387	-8.884313	-9.442656
7.	6	-11.58745	-9.075697	-10.25464
8.	7	-9.260947	-8.631834	-8.92712
9.	8	-9.583155	-7.552726	-8.505745
10.	9	-4.863534	-5.838371	-5.380814
11.	10	-2.026347	-3.48877	-2.802355


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horizon (years)
estimates
polynomial fit
shrinkage estimator
68% pointwise band

Shrinkage: confidence band

- The shrinkage estimator has lower MSE than the original estimator in many cases (higher bias but lower variance).
- It's tempting to try to develop a confidence band that is centered at the shrinkage estimator. Unfortunately, this is very challenging, as any such band would need to take into account the bias in the shrinkage estimator.
- For this reason, I recommend reporting the standard confidence band. The shrinkage estimator should only be viewed as a "best guess" or visual aid.

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Small-sample bias

- We have seen that the LP estimator is unbiased in large samples under weak conditions.
- However, when the data is highly persistent (e.g., when trending variables are included), LP can suffer from substantial bias in finite samples of typical size $T \approx 100$.
- Usually the direction of the bias is such that we under-estimate the persistence of the shock's causal effects on the outcome.
  - Qualitatively similar to small-sample bias of VARs.
- Herbst & Johanssen (2024) develop an analytical bias correction procedure.
  - The formula is not too complicated, but it would be annoying to implement "by hand" in Stata.

Bias correction by differencing

- Piger & Stockwell (2025) propose a simpler way to bias-correct in the special case where we directly observe a shock ε_t.
- They propose to run the LP on first-differenced data. This makes the variables in the regression less persistent, so biases are smaller.
  - We do not difference the observed shock/proxy, since it's already serially uncorrelated.
- To report the impulse responses of the original outcome in levels, we can cumulate the differenced impulse responses.
- Equivalently, we can run an LP with the outcome in long differences:

$$\underbrace{y_{t+h} - y_{t-1}}_{\text{long difference}} = \mu_h + \theta_h \varepsilon_t + \underbrace{\delta'_{h,1}\Delta w_{t-1} + \cdots + \delta'_{h,p}\Delta w_{t-p}}_{\text{lagged first differences}} + \xi_{h,t+h}.$$

Bias correction: discussion

- The Piger & Stockwell approach is simple.
- But it can only be applied when we directly observe a serially uncorrelated shock (or serially uncorrelated proxy thereof).
- Another drawback is that differencing the control variables can sacrifice efficiency relative to LP in levels.
- The Herbst & Johanssen procedure is somewhat more complicated, but can be applied to general LP specifications and does not sacrifice efficiency in large samples.
  - In small samples, the bias correction often comes at the cost of an increase in variance.

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Bootstrap inference

- The basic LP confidence band we've discussed may under-cover at long horizons when the data is highly persistent.
- This can be fixed by the Piger & Stockwell approach when applicable, with possible loss of efficiency.
- Alternatively, we can improve the finite-sample performance by reporting bootstrap confidence intervals for LP.

Bootstrap inference

- Montiel Olea et al. (2025) recommend a VAR-based bootstrap, based on the idea that a VAR is probably a pretty good fit to the data, though imperfect:
  1. Estimate an auxiliary VAR on the real data.
  2. Generate fake data from the estimated VAR.
  3. Apply the LP estimator to the fake data.
  4. Repeat steps 2–3 many times.
  5. Use bootstrapped LP estimates to compute bootstrap confidence interval.
     - See the paper for details.
- Matlab code on GitHub: `https://github.com/ckwolf92/lp_var_nberma`
- Unfortunately not easy to implement directly in Stata.

Bootstrap inference: discussion

- We can also use the bootstrap draws to bias-correct the LP estimator.
- In simulations calibrated to economic data, the bootstrap confidence band has more accurate coverage than the conventional band.
- But the difference is only meaningful at long horizons and for persistent data.
- It's a very good idea to run the bootstrap if you can, at least as a robustness check.

What did we discuss?

- Joint testing by stacking regression results, simultaneous confidence band.
- Smoothing out the estimated impulse response function.
  - Parametric fit.
  - Shrinkage estimator.
  - Report original confidence band.
- Small-sample bias, bias correction.
- Bootstrap inference.